Python Starter Cheat Sheet

Variables, loops, functions and common errors | Ages 10+

Python is read top to bottom, one line at a time. **Indentation matters**: use 4 spaces inside if, for, while and def. Lines starting with # are notes for humans and are ignored by Python.

1. PRINT, VARIABLES AND INPUT	
`print("Hello, world!")`	Shows text on the screen.
`name = "Asha"` `age = 11`	A variable is a named box that stores a value.
`print("Hi", name)`	Print several things, separated by commas.
`print(f"{name} is {age}")`	f-string: puts variables inside text.
`name = input("Your name? ")`	Asks the user to type something (always text).
`age = int(input("Age? "))`	int() turns text into a whole number.

2. DATA TYPES	
`7  -3`	int: whole numbers
`3.14  0.5`	float: decimal numbers
`"hello"  'bee'`	str: text, always in quotes
`True  False`	bool: yes/no values (capital first letter)
`type(age)`	Tells you what type a value is.

3. MATHS AND COMPARING	
`+  -  *  /`	Add, subtract, multiply, divide.
`//  %  **`	Whole-number divide, remainder, power.
`10 % 3  ->  1`	% is handy: n % 2 == 0 means n is even.
`==  !=  >  <  >=  <=`	Compare. Note == (compare) vs = (store).
`and  or  not`	Combine conditions.

4. MAKING DECISIONS	

`if score > 10:`
`  print("Win!")`
`elif score > 5:`
`  print("Nearly")`
`else:`
`  print("Try again")`
	Python checks from the top and runs the first block that is true. Don't forget the colon : at the end of each line.

5. LOOPS	
`for i in range(5):` `  print(i)`	Repeats 5 times: i is 0,1,2,3,4.
`for i in range(1, 6):`	Counts 1 to 5 (stops before 6).
`for fruit in fruits:` `  print(fruit)`	Goes through every item in a list.
`while lives > 0:` `  lives = lives - 1`	Repeats while the condition is true.
`break  continue`	Stop the loop / skip to the next round.

6. LISTS AND DICTIONARIES	
`fruits = ["apple", "mango"]`	A list holds many items in order.
`fruits[0]  fruits[-1]`	First item / last item. Counting starts at 0!

`fruits.append("kiwi")`
`fruits.remove("apple")`
`len(fruits)`
	Add, remove, count items.

`pet = {"name": "Bo", "age": 3}` `pet["name"]`	A dictionary stores key: value pairs.

7. FUNCTIONS	

`def greet(name):`
`  return "Hi " + name`
 	
`print(greet("Asha"))`
	def makes your own reusable command. return sends a value back. Call it by name with brackets.

8. COMMON ERRORS AND WHAT THEY MEAN	
`SyntaxError`	Typo: missing colon, bracket or quote. Check the line above too.
`IndentationError`	Spaces are wrong. Use 4 spaces inside blocks.
`NameError`	Variable not defined yet, or spelled differently (Score vs score).
`TypeError`	Mixing types, e.g. "5" + 3. Use int() or str().
`IndexError`	List position doesn't exist. Last index is len(list) - 1.
`Infinite loop`	while never becomes false. Press Ctrl + C to stop.

9. HANDY BUILT-INS	
`len("bee")  len(fruits)`	How many characters or items.
`"hello".upper()  .lower()  .title()`	Change the case of text.
`"a-b-c".split("-")`	Chop text into a list: ["a","b","c"].
`round(3.14159, 2)  abs(-5)`	Round a number / remove the minus sign.

`import random`
`random.randint(1, 6)`
`random.choice(fruits)`
	Random number (a dice roll) / random pick.

`import time` `time.sleep(1)`	Pause for 1 second.
`sorted(nums)  max(nums)  sum(nums)`	Sort, biggest, total of a list.

MINI PROJECT: Guess the Number

```
import random
secret = random.randint(1, 20)
guess = 0
tries = 0
while guess != secret:
    guess = int(input("Guess 1-20: "))
    tries = tries + 1
    if guess < secret:
        print("Too low")
    elif guess > secret:
        print("Too high")
print(f"You got it in {tries} tries!")
```

Level up: limit the player to 5 tries, let them choose the range, or add a score that goes up when they win.

Where to practise (free): trinket.io, replit.com or python.org/shell. Try changing one thing at a time and predict what will happen before you run it.